TwilioConference

Genesis AudioDevice Logic – Audio Routing Modes

Mode Selection Based on ACTIVE/INACTIVE Flags

Audio routing behavior is determined by the **ACTIVE** status of `GenesisAudioDevices` at a **site** (location or logical group). There are **three modes**, defined as follows:

1. 🏢 Conference Mode – *Full Collaboration*

- **Condition**:
  All `GenesisAudioDevices` at the site are marked as **ACTIVE**.
- **Behavior**:
  - Activates **Conference Call Mode**.
  - Devices **share audio** through a central **conference bridge**.
  - Suitable for **multi-device**, **multi-user**, or **team-based** environments.
- **Use Case**:
  Modern conference rooms with multiple devices needing synchronized audio participation.

2. 📞 Dial-In Mode – *Single Device Control*

- **Condition**:
  Exactly one `GenesisAudioDevice` at the site is **ACTIVE**, and **all others are INACTIVE**.
- **Behavior**:
  - Engages **Dial-In Behavior**.
  - The ACTIVE device **links directly to the audio sensor**.
  - **No linking logic** for INACTIVE devices — they're ignored.
- **Use Case**:
  Legacy or simplified setups where a single device manages the audio (e.g., personal office dial-in).

3. 🎙️ Device Audio Mode – *Isolated Operation*

- **Condition**:
  All `GenesisAudioDevices` at the site are **INACTIVE**.
- **Behavior**:
  - Devices **use their own internal audio** (microphone/speaker).
  - **No audio sharing or linking**.
  - Simplest mode – no dependencies.
- **Use Case**:
  Fallback operation or independent device use.
  ⚠️ **Note**: This behavior is pending confirmation in production.

Improved Audio Routing with Conference Mode Support

Previous Behavior:

- When multiple **Genesis audio devices** were configured under a single site, initiating an audio call from the UI (for any camera under that site) would always **route audio to the first added Genesis audio device**, regardless of its status.

New Behavior:

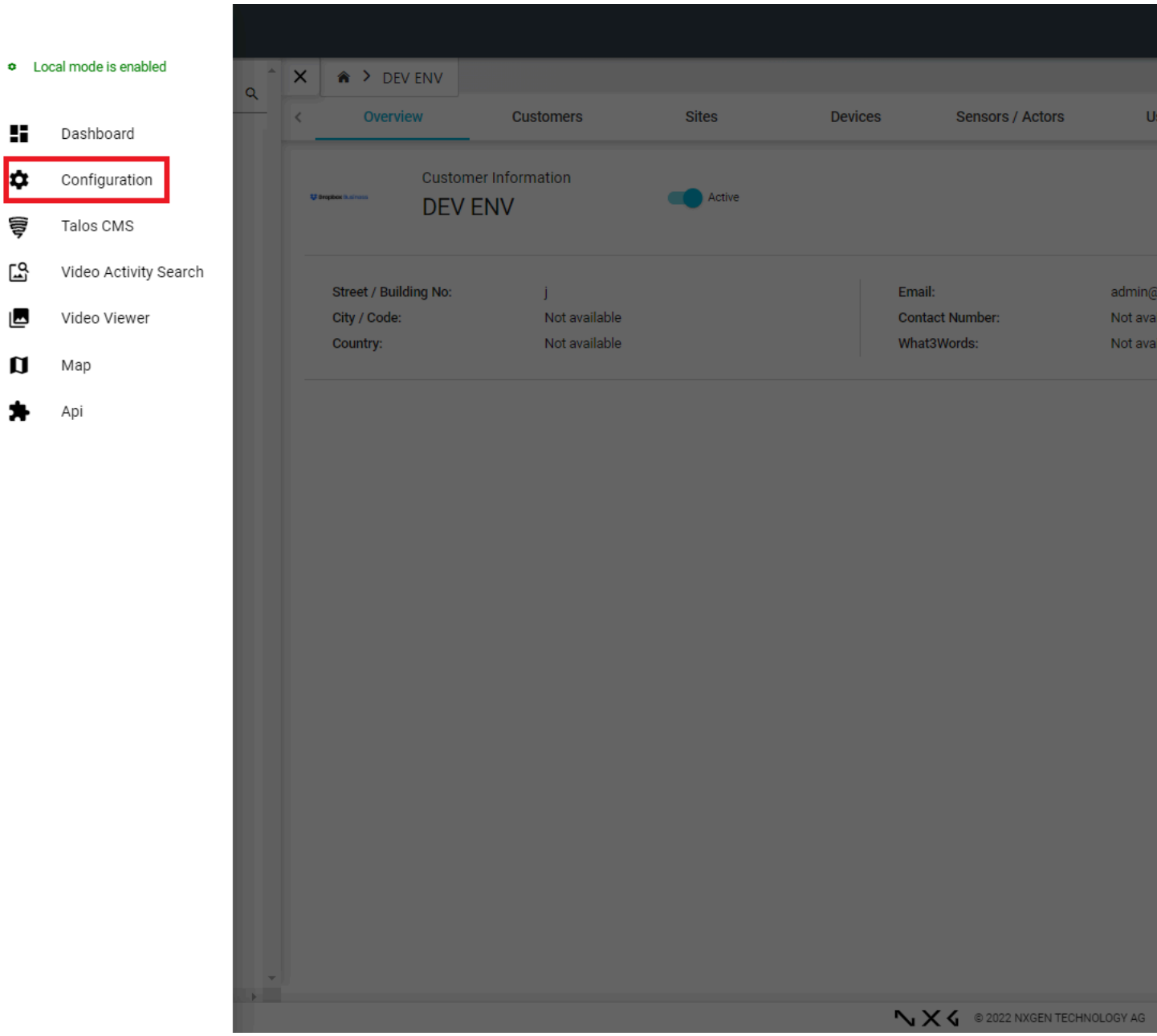
- When an audio call is initiated and there are **multiple active Genesis audio devices** under the same site, the system will now initiate a **conference mode**.
- In this mode:
  - **All active Genesis audio devices** and the **Genesis-side operator** will be **connected in a conference**.
  - This ensures collaborative communication across all active audio endpoints within the site.

Fallback Handling:

- If **none of the Genesis audio devices are active**, or if the **site has no Genesis audio devices configured**:
  - The call will fall back to using the **device's default audio**, **if supported** by that particular device type.
  - **NOTE:**
    - Genesis audio is configurable only if the user has the permission
    - "Local mode" settings are not required (calls can be initiated via Genesis audio even if the local mode is disabled)

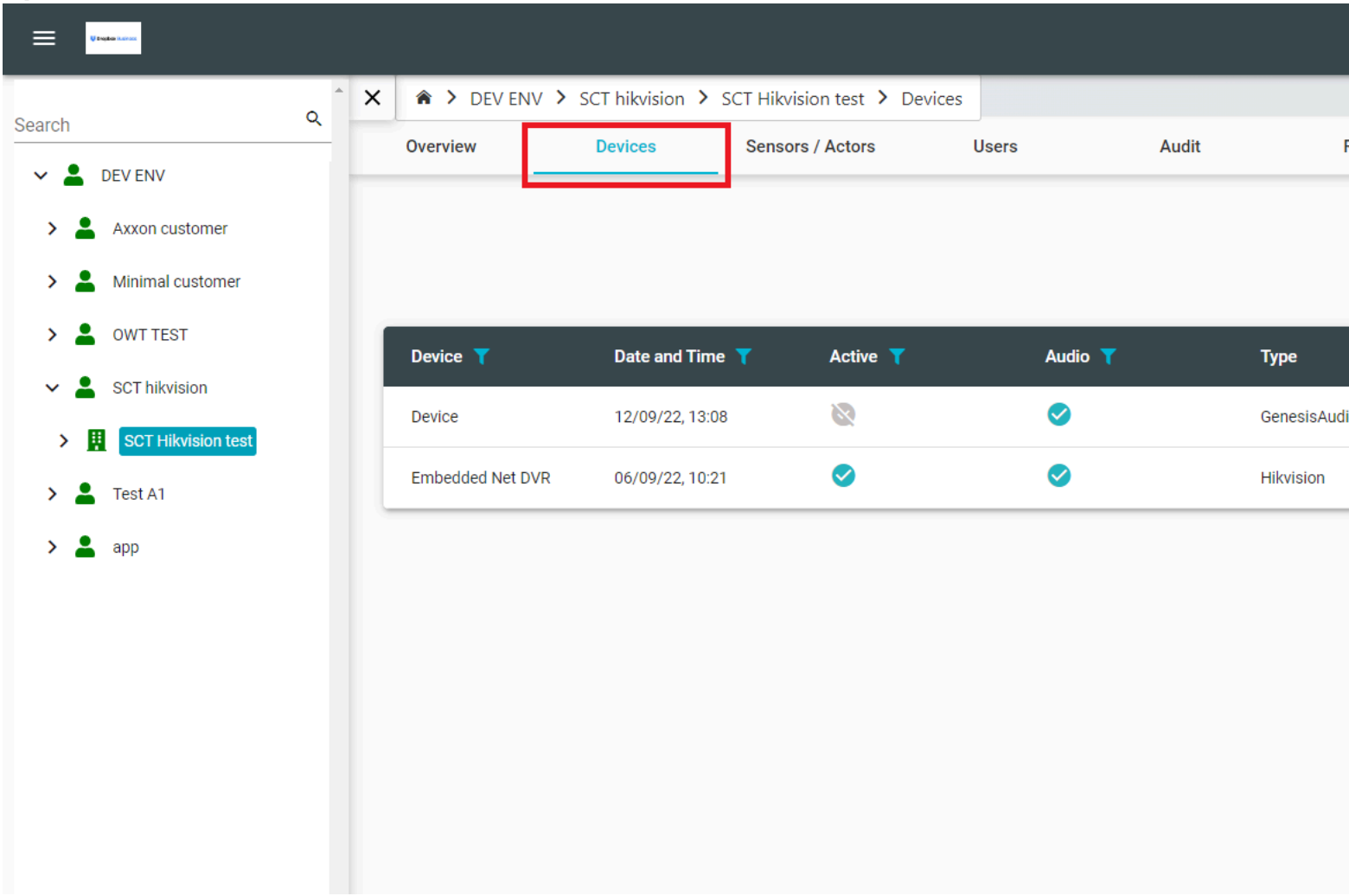
Steps to add an Genesis audio device under the site:

Step 1: Login to Genesis
Step 2: Navigate to the Configuration app in Genesis

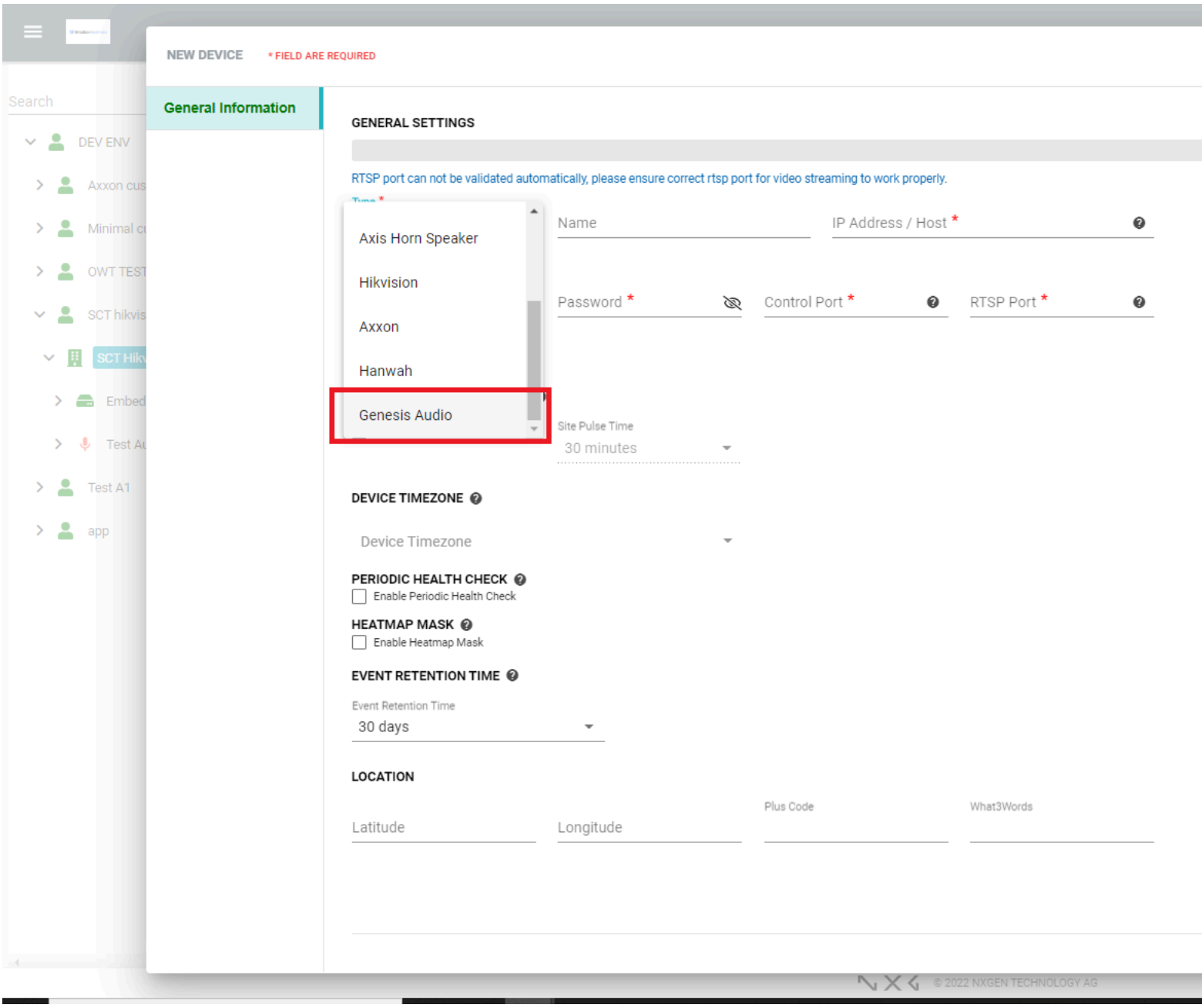


Step 3: Select the Site under which the Genesis audio device is need to be added
Step 4: Under the selected site, navigate to the "Devices" tab

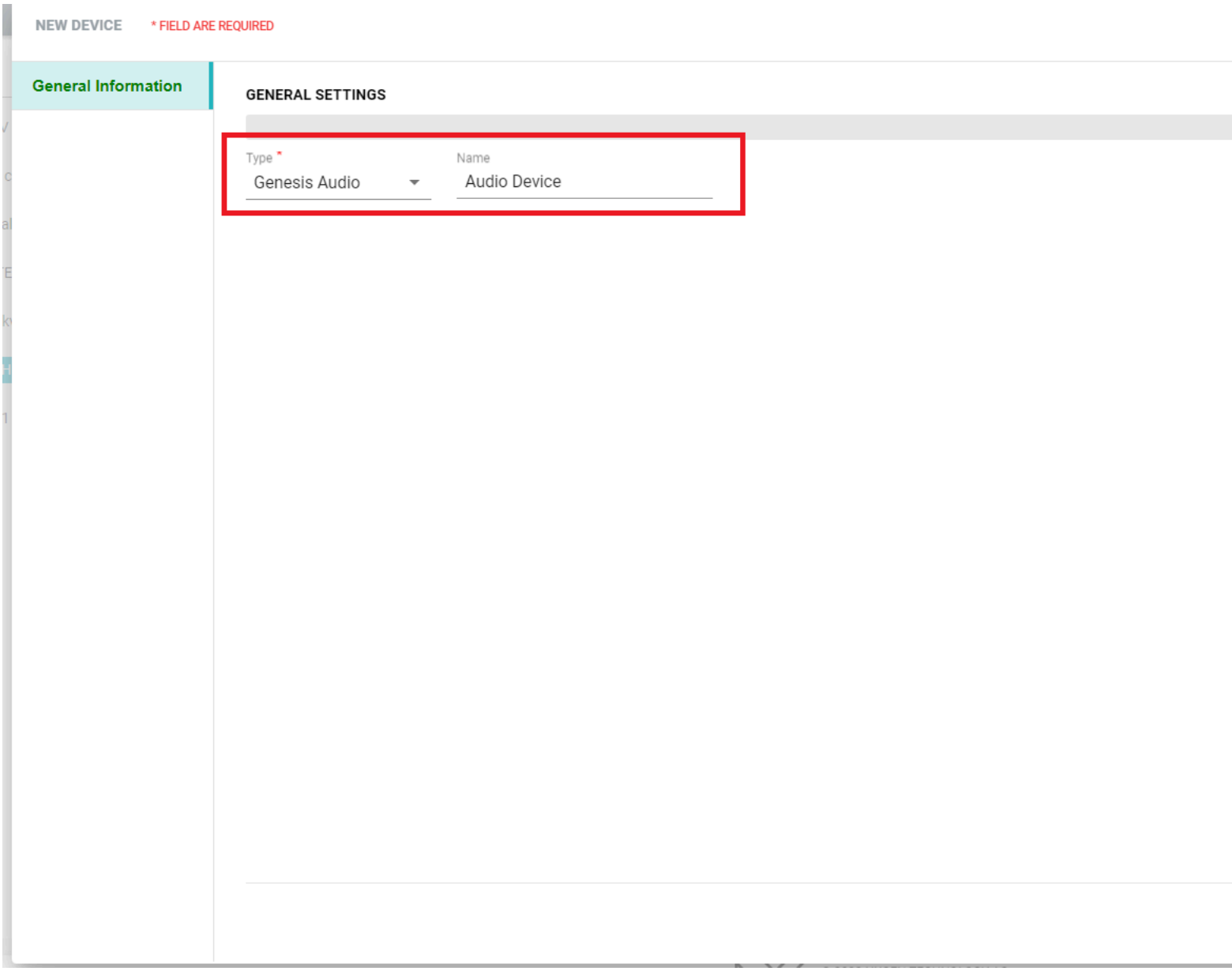
Step 5: Click on the "Add" button



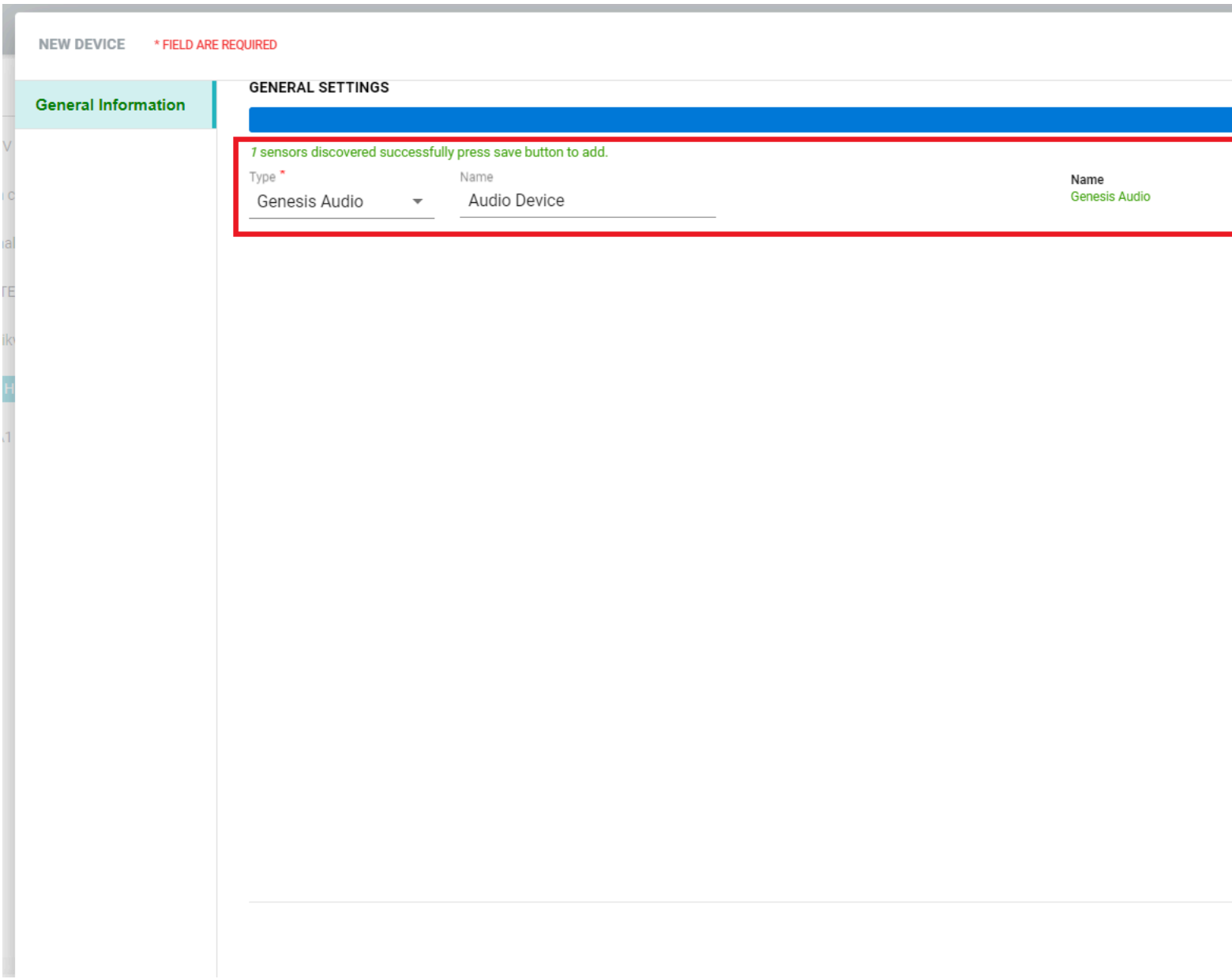
Step 6: Under General Settings, click on "Type" which gives the list of devices that are available. Select "Genesis Audio"



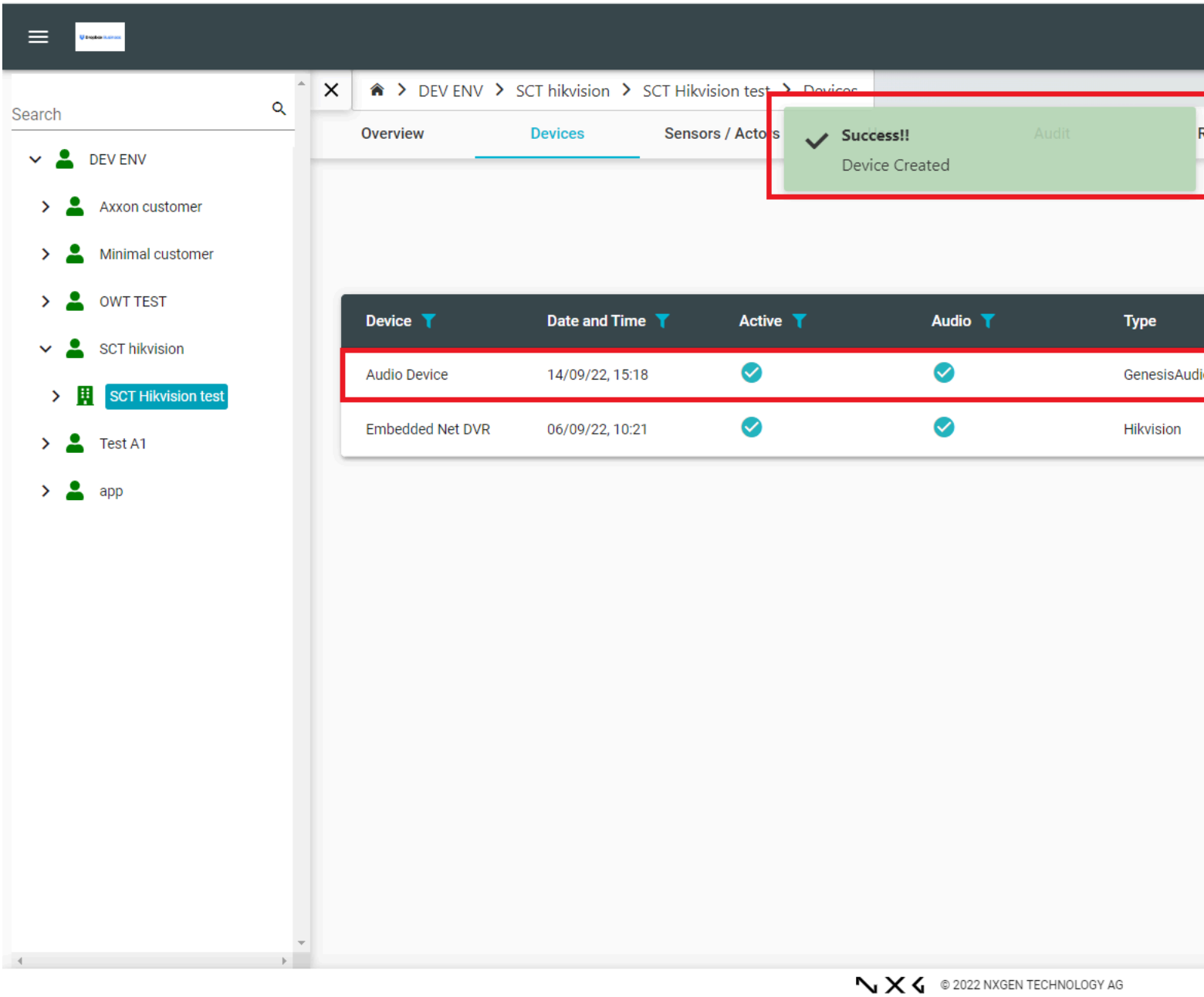
Step 7: Enter the desired name and click on "Discover"



Step 8: Once the device is discovered, click on "Save"



Step 9: A success pop up message "Device Created" will be displayed and the device will be added to the data table



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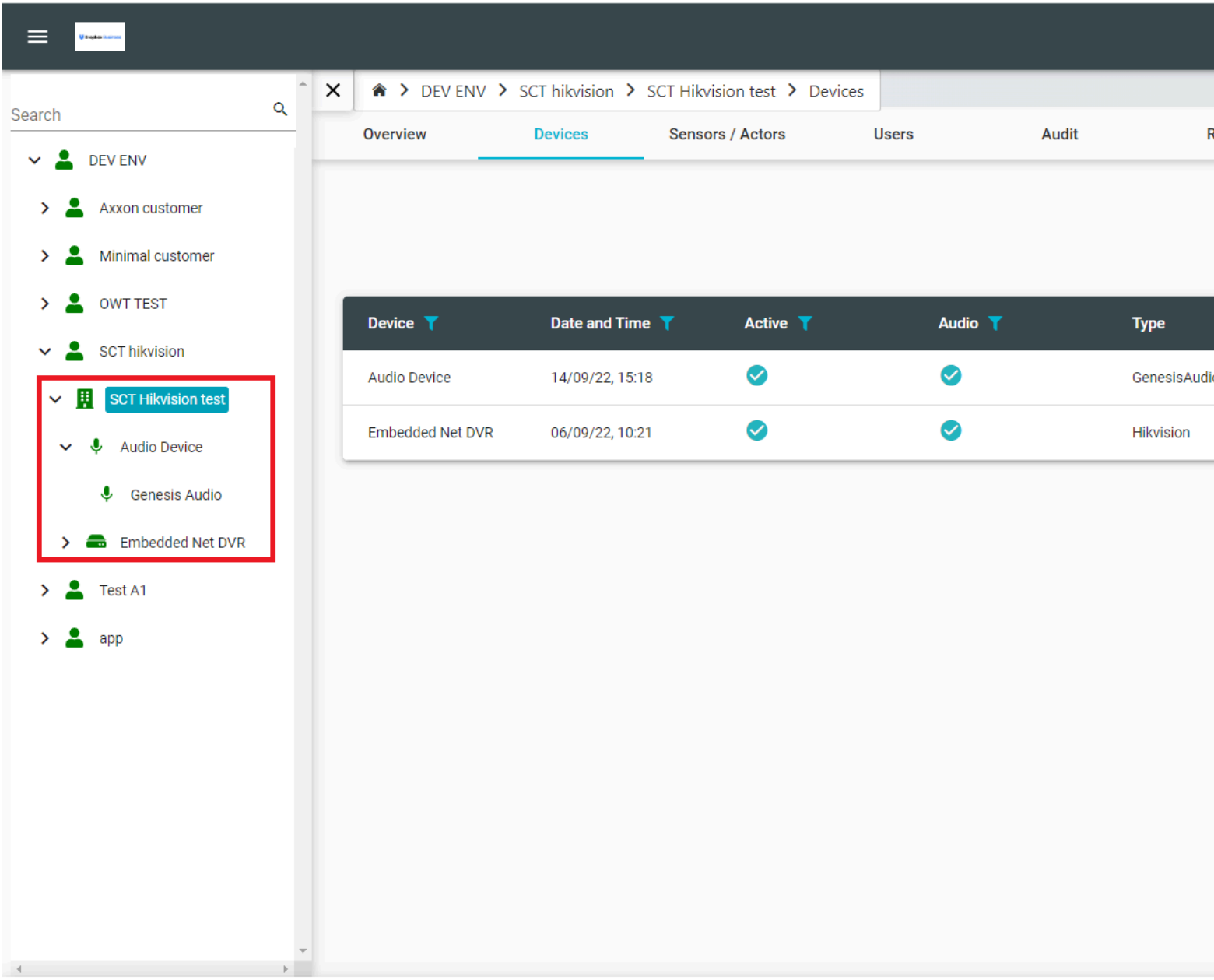
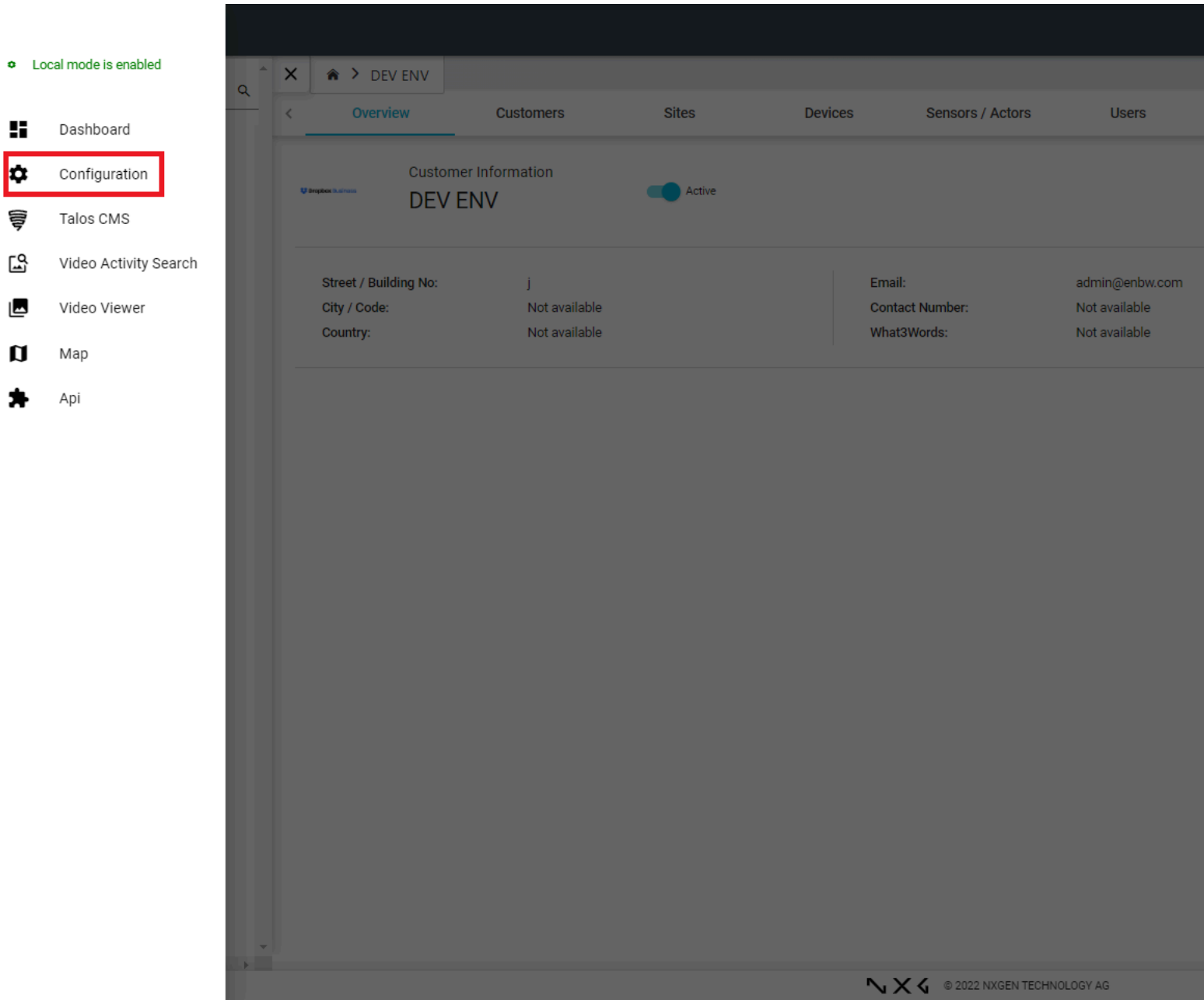
Step 10: Verify if the added audio device is added by expanding the treeview under the site



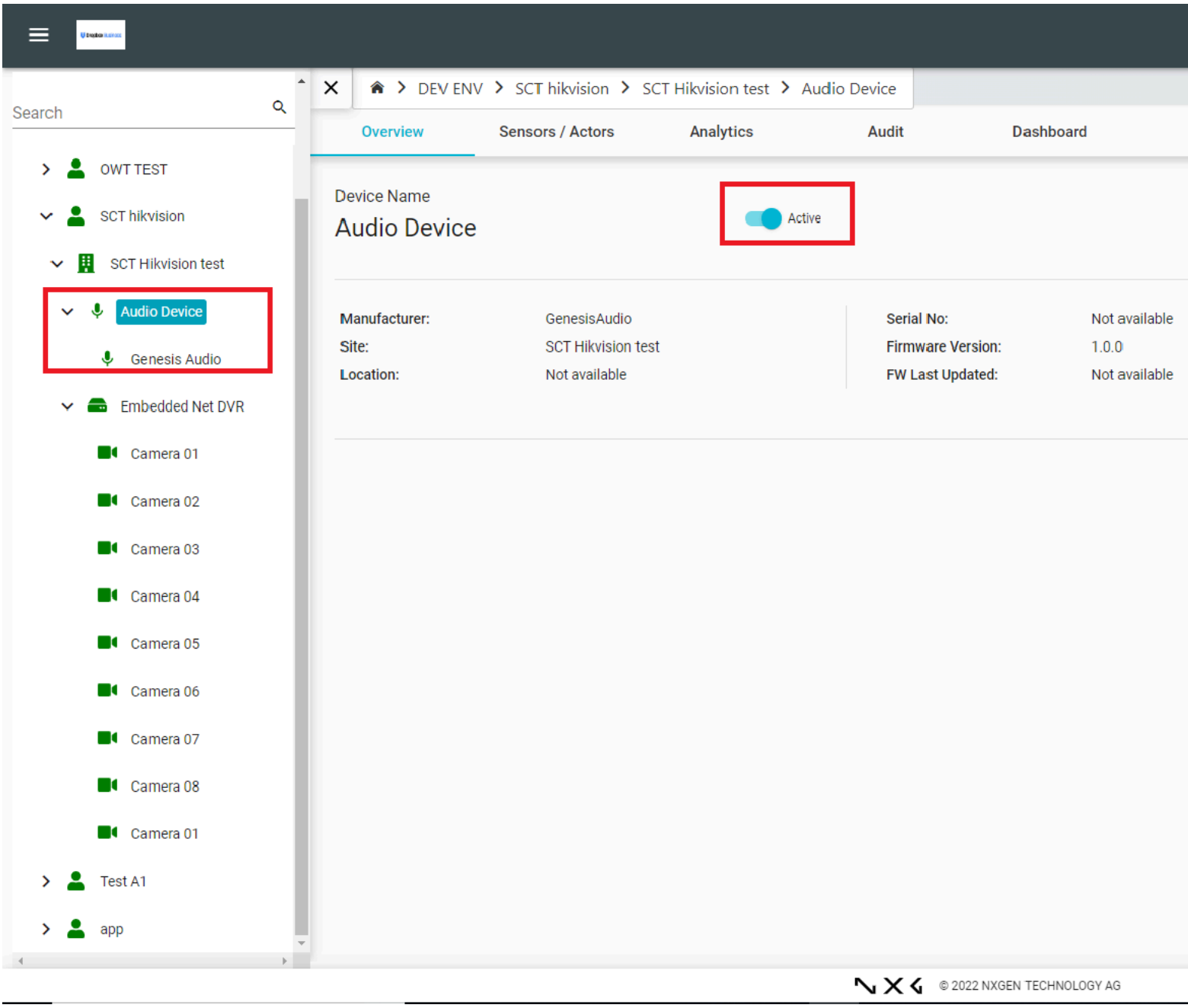
NOTE: Use this Genesis Audio device credentials and configure the IP speaker.

Steps to initiate an Genesis Audio call request from the Salvo:

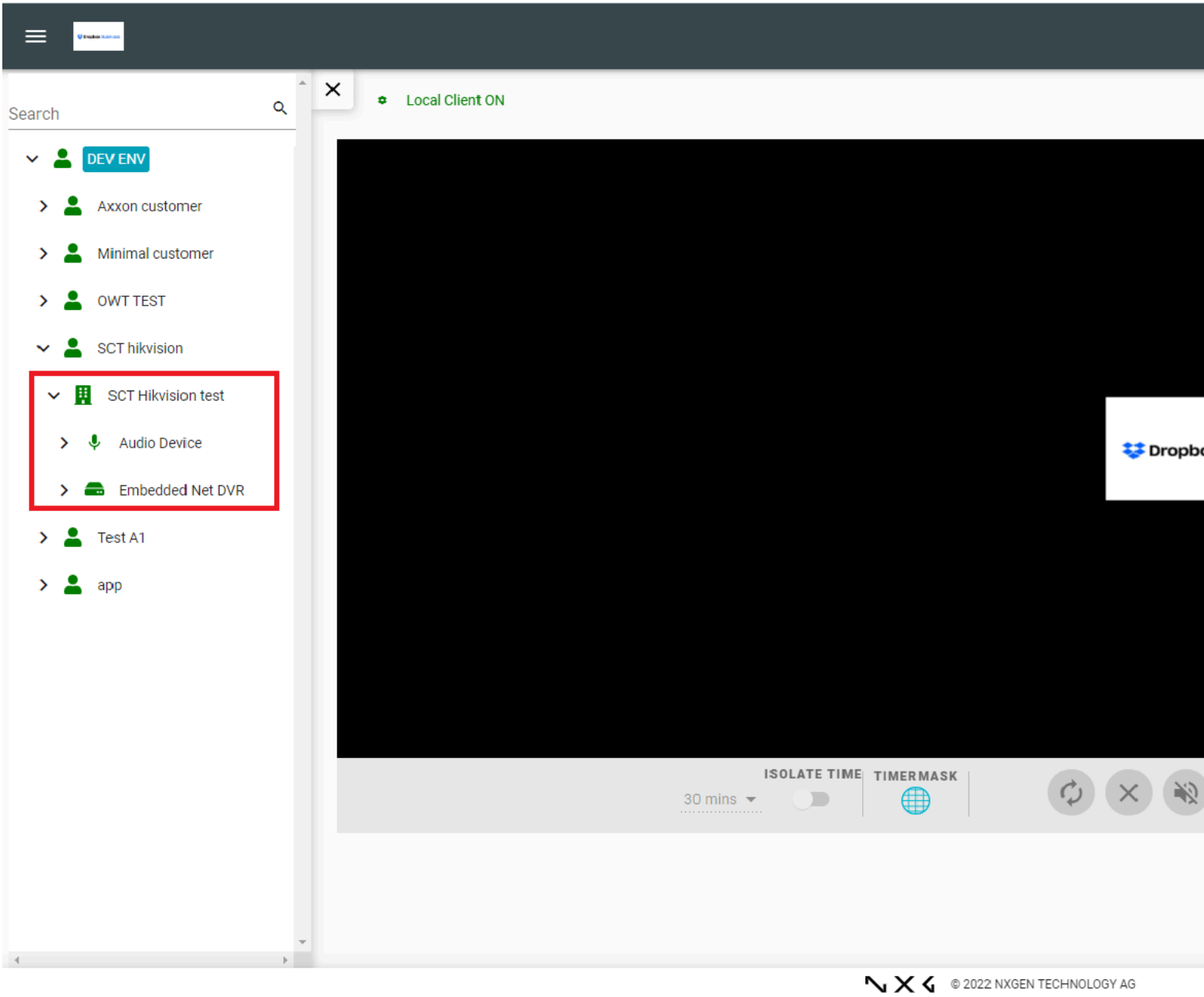
Step 1: Login to the Genesis website
Step 2: Navigate to the Configuration app in the Genesis



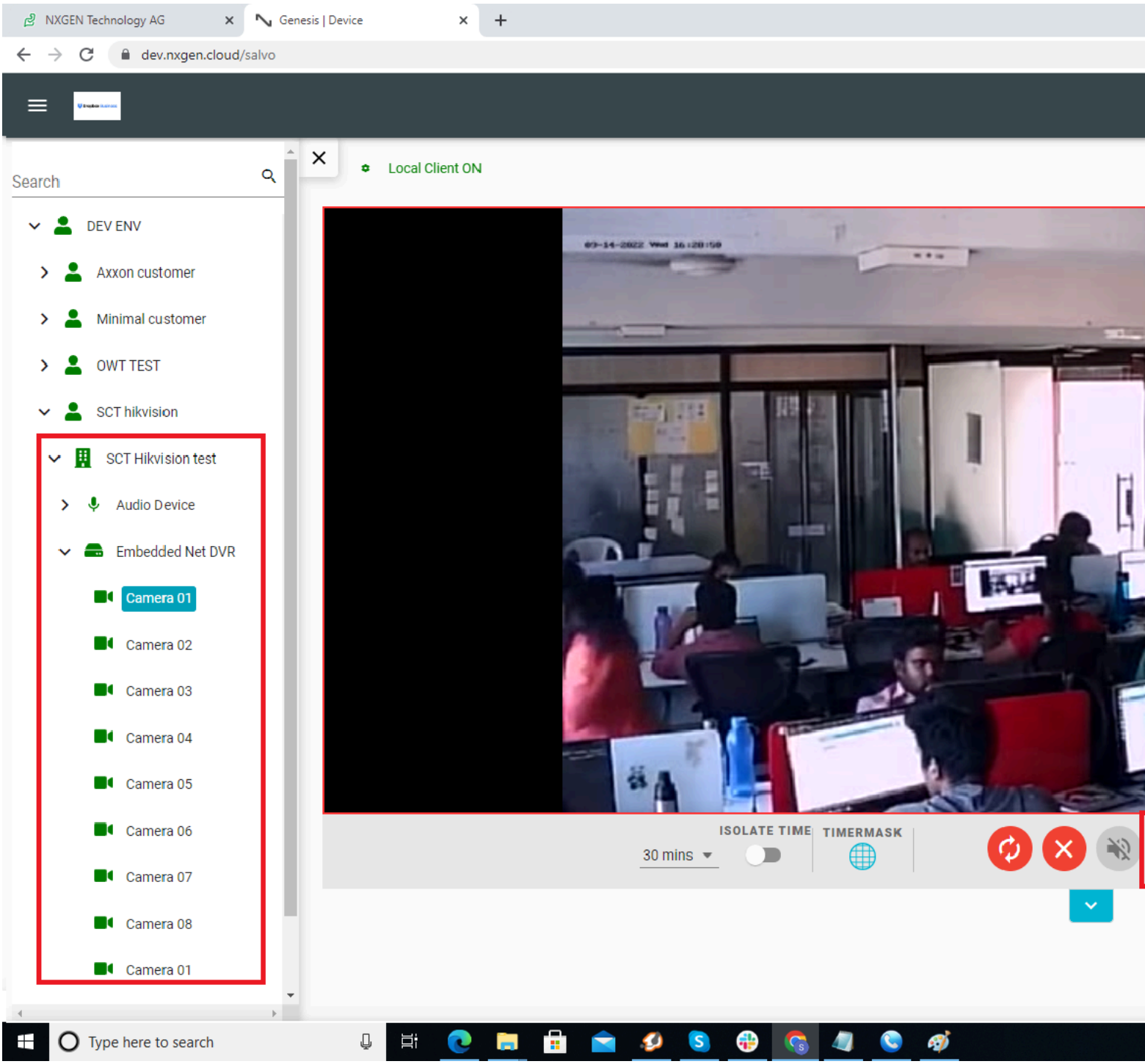
Step 3: Select the site in the tree view and right click to Expand the site
Step 4: Select the added Genesis audio device under the site in the tree view
Step 5: Verify if the "Active" toggle is enabled to indicate that the added Genesis audio device is active



Step 6: Navigate to the Video viewer (Salvo) app
Step 7: Select the site in the tree view and expand the site
Step 8: Verify if the Genesis audio device is added under the selected site

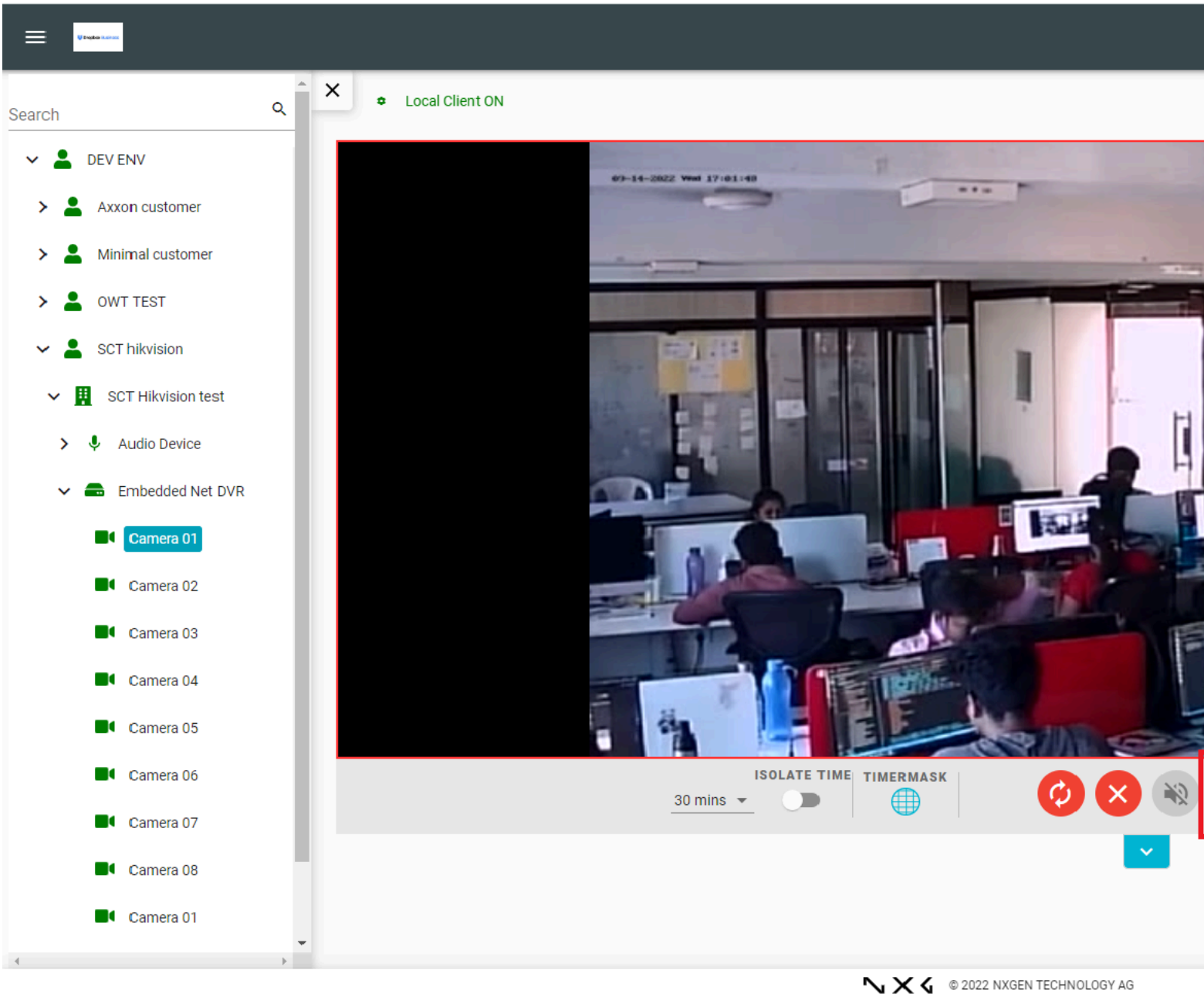


Step 9: Now drag and drop an sensor from the site into the video viewer (Salvo)
Step 10: Verify the mic symbol is enabled (red colour).



Step 11: Click on the mic symbol, an audio call will be initiated

Step 12: Once the call is connected, the color of the mic will be changed from red to blue



Step 13: Click again on the mic to disconnect the call. Color of the mic will change from blue to red, once the call is disconnected.